Supplementary Material

Narrative Agency under Evidence Constraints

S1. Related-work positioning

This table reproduces the related-work comparison moved from the main manuscript. It makes explicit how the framework differs from adjacent formulations and what additional objects it represents.

Table S1: Positioning relative to adjacent problem formulations.

Prior line	Primary object	What it naturally expresses	What this framework adds
Abduction / plan recognition	Hypotheses or latent plans	Rank or infer explanations from observations	An objective-invariant replay-admissible history set, followed by agent-specific selection
Narrative planning	Intentional plot structure	Construct a coherent story or plan toward a goal	Preserve every evidence-compatible causal history before choosing a report
Strategic communication	Messages, beliefs, or equilibria	Model persuasion, signaling, or deception incentives	Separate physical feasibility from self-interested testimony and interactive corroboration
This work	Feasible histories plus J_i	Select incompatible accounts from one shared record	Express cases where each account is individually feasible, yet changing objectives or social pressure changes only the selected account

S2. Interactive testimony outcome summary

This table contains the outcome summary omitted from the main manuscript to make room for the more directly interpretable round-by-round hearing trace. The public record is the fixed *In a Grove* "run-06" record. The simulator, utilities, and feasible pool remain fixed while the interaction parameters vary. The "distinct" count is the number of terminal causal claims. The "incompatible" count is the number of claims that cannot coexist with the joint profile under the record. The symbol $\tau^\star$ denotes the simulator's reference history.

Table S2: Outcome summary for interactive testimony on the fixed *In a Grove* "run-06" record.

condition	revisions	distinct	incompatible	$\tau^\star$ retained?
Independent ($\beta = 0$)	0	3	2	yes
Corroborative ($\beta = .5, \gamma = 0$)	2	1	0	no
+ accountability ($\beta = .5, \gamma = .2$)	0	3	2	yes

The corroborative condition collapses the three opening accounts to one and removes the two jointly incompatible claims. Accountability prevents those revisions by penalizing drift from the openings and restores the three-way profile.

S3. Replay-learning curves

Figure S1 reproduces the Ring replay-learning curves reported as Figure 2 in the main manuscript. Each line is the median best-feasible utility gap over eight seeds, with shading for the interquartile range. A gap of zero is the exact enumerated optimum.

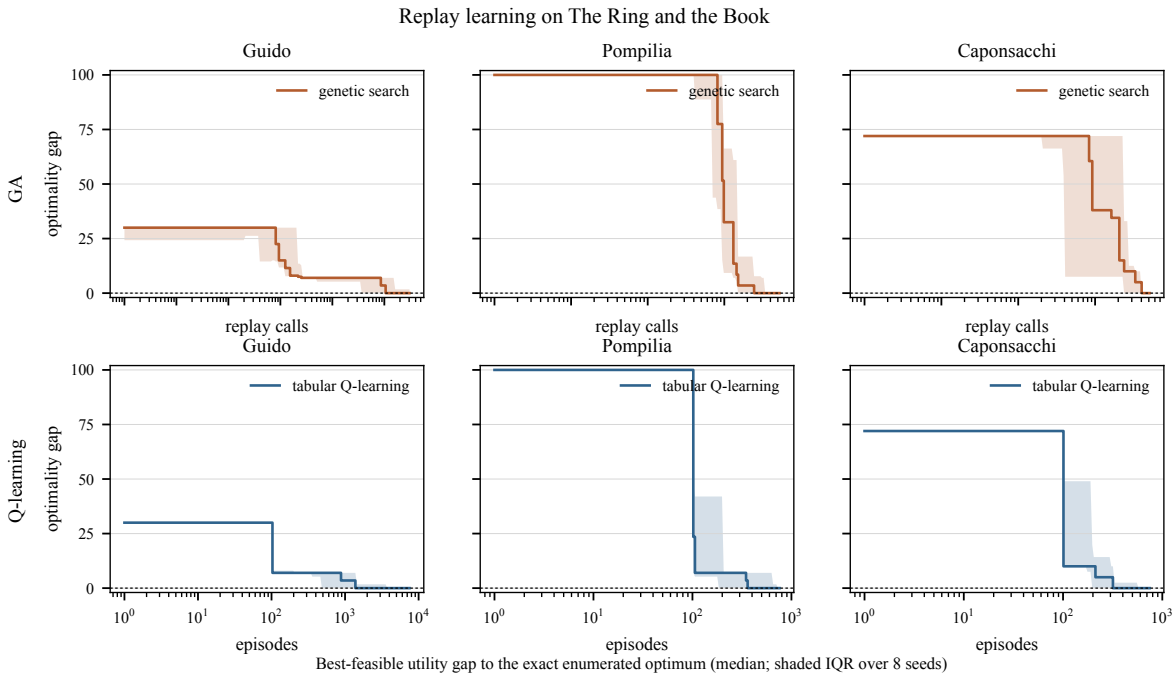


Figure S1: Empirical replay-learning curves on *The Ring and the Book*. The top row shows genetic search against replay-call budget and the bottom row tabular Q-learning against episode budget. The curves are empirical learning trajectories, not a formal convergence guarantee.

S4. Full prompting ladder and deterministic baselines

This table gives the complete version of Table 2 from the main manuscript. The three primary generator models use the same samples and conditions as the main experiment. GPT-5 and Claude Opus 4.8 are separate three-sample probes. Solver rows report their stated run or exact reference. †The account is excluded by F. ‡Distinct counts the three beam outputs.

Table S3: Full prompting ladder and deterministic baselines on *In a Grove*.

	utility attained				
Solver / condition	bandit	wife	samurai	infeasible	distinct
Not enforced (arg max over all assertions)	72†	75	80	23.2%	3.0
zai-glm-4.7 (instructed)	48	43	80	11.1%	3.0
+ physical entailments spelled out	48	43	80	0%	3.0
+ self-check	48	43	80	0%	3.0
+ self-revise	48	43	80	0%	3.0
gemma-4-31b (instructed)	48	43	80	0%	3.0
+ physical entailments spelled out	48	43	80	0%	3.0
+ self-check	48	43	80	11.1%	3.0
+ self-revise	48	43	80	11.1%	3.0
gpt-oss-120b (instructed)	48	75	42	0%	1.0
+ physical entailments spelled out	48	75	42	0%	1.0
+ self-check	48	43	80	0%	3.0
+ self-revise	48	75	42	0%	1.0
gpt-5 (instructed)	42	43	80	0%	3.0
+ consequence	42	43	80	0%	3.0
+ self-check	42	43	80	0%	3.0
+ self-revise	42	43	80	0%	3.0
Claude Opus 4.8 (instructed)	42	43	80	0%	3.0
+ consequence	42	43	80	0%	3.0
+ self-check	42	43	80	0%	3.0
+ self-revise	42	43	80	11%	3.0
Constrained beam planner ($w = 32$, $T = 7$)	58	75	73	0%	1.0‡
Genetic search (8 seeds)	58	75	80	0%	3.0
Tabular Q-learning (8 seeds)	58	75	80	0%	3.0
Simulator (arg max over enumerated Feasible(F))	58	75	80	0%	3.0

S5. Utility definitions, solver protocol, and hyperparameters

S5.1. Concrete definition of U_i

The utility is computed entirely from the replayed causal signature, before any language realization. For the Grove domain, replay returns six normalized features: agency, honour, clean hands, sympathy, dignity, and wronged. Their deterministic values are computed from the fatal actor, action type, weapon, victim status, and whether the victim was bound or armed. The implementation is frozen in `experiment-code/recon/grove-objective.mjs`.

The witness-specific feature triples are

$$\begin{aligned} U_{\text{Tajomaru}} &= \tfrac{100}{3}(\text{honour} + \text{agency} + \text{cleanHands}), \\ U_{\text{Masago}} &= \tfrac{100}{3}(\text{sympathy} + \text{dignity} + \text{cleanHands}), \\ U_{\text{Takehiro}} &= \tfrac{100}{3}(\text{honour} + \text{agency} + \text{wronged}). \end{aligned}$$

Each component lies in $[0, 1]$, so $U_i \in [0, 100]$. In particular, agency is 1 when the witness deals the fatal blow and 0.15 otherwise. The clean hands score is 0.15 for the fatal striker and 1 otherwise. Dignity is $0.4 + 0.4\,\text{agency}$. Honour is 1 for Takehiro's self-strike, 0.6 for an impalement on a held weapon, 0.05 when Takehiro is bound at death, 0.5 when he is armed, and 0.3 otherwise. For any other witness, honour is 1 when that witness makes an armed longsword strike, 0.1 when striking a bound victim, 0.3 for another strike, and 0.6 when not the fatal actor. Sympathy is 0.35 for the fatal striker, 0.7 for Takehiro when not the striker, and 0.8 otherwise. Wronged is 0.8 for Takehiro when another witness strikes, 0.4 for his self-strike, 0.6 for his other outcomes, and 0.3 for the other witnesses. These rules contain no literary labels. Those descriptions are consequences of the selected causal signature.

For the agent-operations domain, replay computes six features from the actor, decision owner, plan author, authorization channel, intent, and monitor flag: clean hands, non-decision, authorization, deviation, vigilance, and vindication. The fixed triples are Executor (authorization, non-decision, clean hands), Planner (deviation, clean hands, non-decision), and Monitor (vigilance, vindication, clean hands). For *The Ring and the Book*, the six features are warrant, provocation, bearing, innocence, purity, and efficacy. The fixed triples are Guido (warrant, provocation, bearing), Pompilia (innocence, provocation, purity), and Caponsacchi (purity, efficacy, innocence). The branch values are deterministic functions of replayed causal fields and are listed in `ops-objective.mjs` and `ring-objective.mjs`. No learned parameters or language-model scores enter any U_i.

All non-enumerative solvers interact with a domain only through legal-action generation, replay, evidence-miss counts, and utility evaluation. The reported seeds are $\{1, \ldots, 8\}$, using the deterministic seeded generator in the released code. A run stops early when it reaches the exact enumerated utility. Otherwise, it stops at the stated budget.

Algorithm S1 Replay-based best-response solvers for $\hat{\tau}_i$ under J_i

(a) Genetic search

Require: simulator, F, fixed profile ρ_{-i}, opening account τ_i^0, objective J_i

$P \leftarrow$ random legal histories, $A \leftarrow \emptyset$
for each generation **do**
  for all $\tau_i \in P$ **do**
    $m \leftarrow \text{misses}(\text{replay}(\tau_i), F)$
    score τ_i by $(m, -J_i(\tau_i, \rho_{-i}, \tau_i^0), |\tau_i|)$
    add τ_i to A if $m = 0$
  end for
  $P \leftarrow$ select/recombine/mutate/repair
  locally perturb the incumbent
end for
return $\arg\max_{\tau_i \in A} J_i(\tau_i, \rho_{-i}, \tau_i^0)$

(b) Tabular Q-learning

Require: simulator, F, fixed profile ρ_{-i}, opening account τ_i^0, objective J_i, horizon T

$Q \leftarrow 0$, $A \leftarrow \emptyset$
for each episode **do**
  reset simulator, $\tau_i \leftarrow$ empty history
  for $t = 1$ to T **do**
    choose legal a by ϵ-greedy Q
    append a to τ_i and step simulator
  end for
  $m \leftarrow \text{misses}(\text{replay}(\tau_i), F)$
  $r \leftarrow J_i(\tau_i, \rho_{-i}, \tau_i^0)$ if $m = 0$, else $-\lambda m$
  update Q backward along τ_i
  add τ_i to A if $m = 0$
end for
return $\arg\max_{\tau_i \in A} J_i(\tau_i, \rho_{-i}, \tau_i^0)$

For genetic search, the primary fitness key is the number of evidence misses, followed by utility and then shorter history length. Crossover is legalised by replay and mutation extends a legal prefix. For Q-learning, a feasible episode receives $1 + \bar{U}_i$ and an infeasible episode receives $-0.2m$, where m is the number of evidence

Table S4: Replay-solver hyperparameters used for the reported recovery results.

solver	domain(s)	horizon	budget	stochastic settings and stopping rule
genetic search	Grove, Ops	$T = 7$	POP = 40, GENS = 40	crossover probability 0.7, mutation probability 0.3, tournament size 3, elitism, and 8 local perturbations per generation. Stop at exact optimum or generation cap
genetic search	Ring	$T = 24$	POP = 80, GENS = 300	same crossover, mutation, tournament, elitism, and local-search settings. Stop at exact optimum or generation cap
tabular Q-learning	Grove, Ops	$T = 7$	6,000 episodes	learning rate $\eta = 0.4$, discount $\gamma_Q = 0.95$, legal-action masking, and linear ϵ schedule $0.90 \rightarrow 0.05$. Stop at exact optimum or episode cap
tabular Q-learning	Ring	$T = 24$	12,000 episodes	same η, γ_Q, action masking, and ϵ schedule. Stop at exact optimum or episode cap
constrained beam	Grove	$T = 7$	width $w = 32$	deterministic decoding with illegal actions masked. No stochastic seed sweep

misses. The cost columns in the main solver table are not the same unit: enumeration counts explored states, genetic search counts simulator calls to first reach the optimum, and Q-learning counts episodes to first reach it.

S6. Ring filtration and sensitivity audit

The Ring candidate space contains 576 distinct causal assertions before the public record is applied. Replaying the fixed eleven-document record leaves 36 feasible assertions, a 93.75% reduction. These are assertion counts. The corresponding enumeration explored 32,928 reachable states. The Ring domain is derived from Browning's account of the Franceschini prosecution. The primary-source citation appears in the main manuscript.

Table S5: Ring assertion-space filtration under the fixed public record.

candidate assertions	feasible assertions	removed	reduction
576	36	540	93.75%

For the Grove sensitivity audit, the 19 hand-set feature constants are on a $[0, 1]$ scale. In each of 5,000 trials (seed 16980222), every constant receives an independent offset drawn uniformly from $[-0.4, 0.4]$ and is then clipped to $[0, 1]$. The simulator and feasible set remain fixed. For each trial we recompute each witness's utility argmax over that feasible set. A trial counts as preserving the full spread when the three selected causal signatures are distinct. This occurs in $4{,}040/5{,}000 = 80.8\%$ of trials, with a 95% Wilson interval of $[79.7, 81.9]\%$. Thus "± 0.4" is an absolute offset on the normalized feature scale, not a relative percentage change.